Voice Command Navigation for the Boston Dynamics Spot Robot

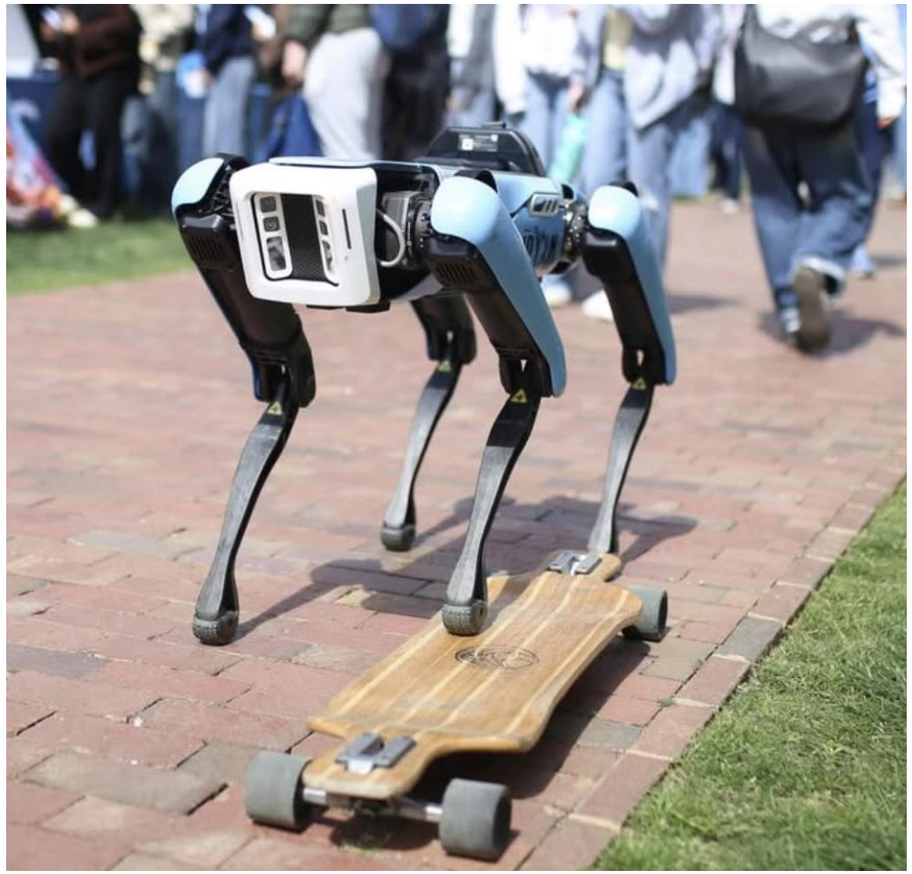
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What is Spot?

Spot is a four-legged robot developed by Boston Dynamics. Because it walks instead of using wheels, it can move through spaces built for people, including hallways, doorways, stairs, and uneven surfaces.

Why Spot?

Spot can already navigate through complex environments on its own. This makes it a useful platform for exploring a bigger question: how can we make controlling a robot as simple and natural as telling it where we want it to go?

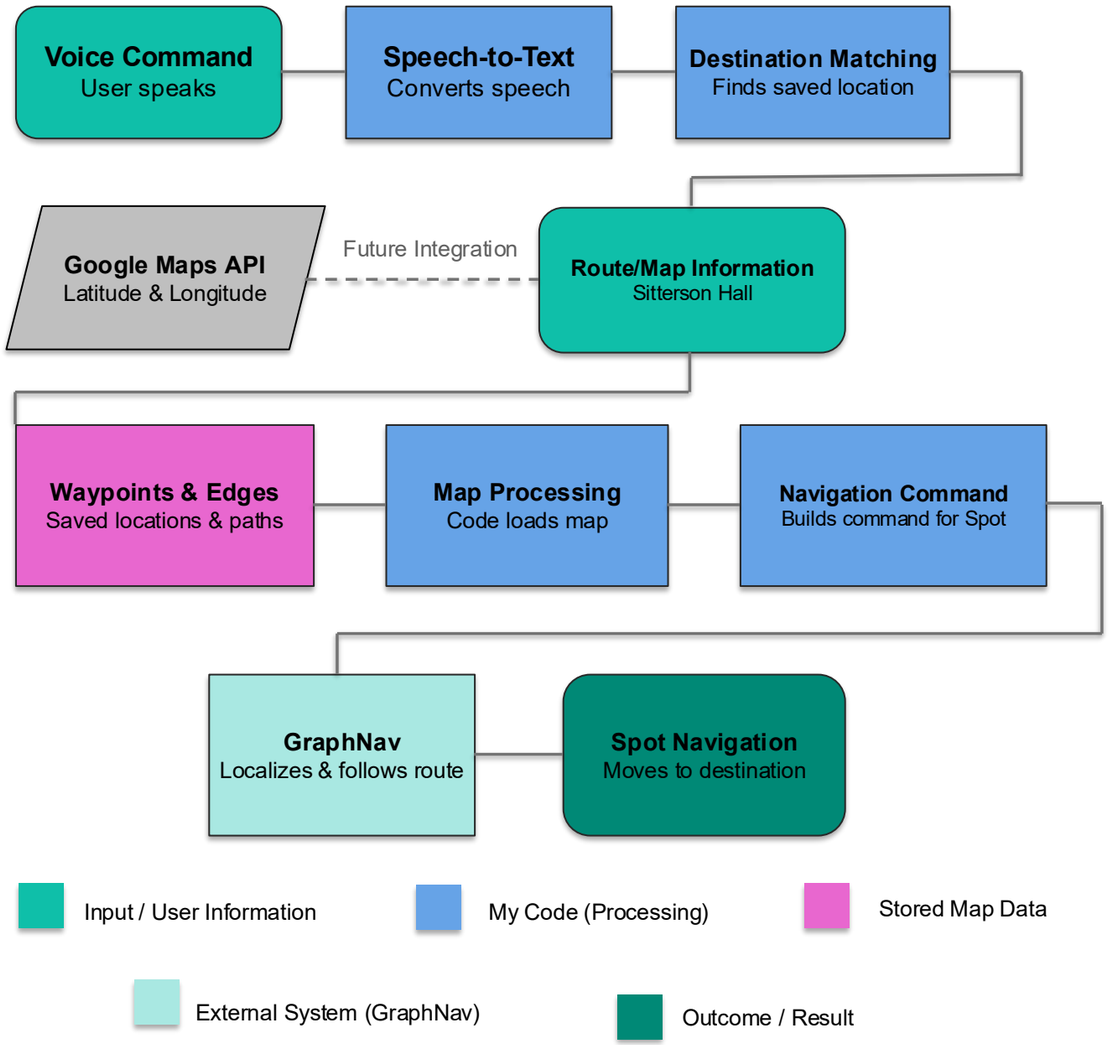




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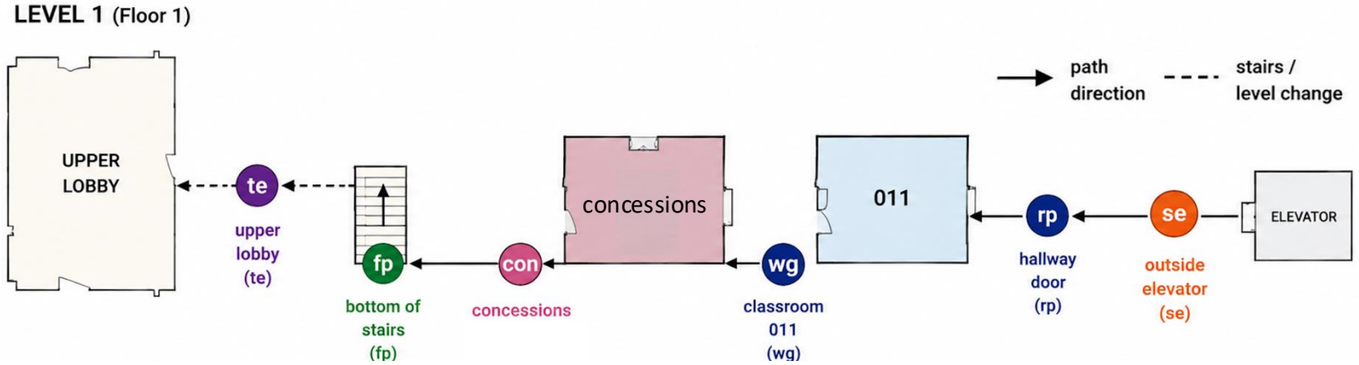
How Can Spot Understand Where We Want It to Go?

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What am I trying to figure out?

How can we translate where a person wants Spot to go into locations and paths that Spot can understand and navigate?



How Spot sees the map

Spot represents the building as waypoints (saved locations) connected by edges (paths it can travel). It uses this network to locate itself and navigate to a destination.

Why does this matter?

Spot and traditional mapping tools represent locations differently. Connecting these systems could eventually allow coordinates and routes from tools like Google Maps to be translated into navigation instructions Spot can understand.

What Were the Results?

What are my results?

I created a working navigation system for Spot using a mapped area inside Sitterson Hall. I built Python code that loads the map, identifies saved locations, connects them through paths, and communicates with Spot's navigation system. I then expanded the system to support voice commands, using a language model to translate natural spoken requests into commands Spot can act on using its navigation system. I also connected the system to an NVIDIA Jetson, working toward running the full system directly on the robot.

Why are these results important to the research community?

This work demonstrates how a language model can translate natural voice commands into instructions that Spot can act on using its internal navigation system. It provides a foundation for making autonomous robots easier and more natural for people to control.

Why are these results important to a general audience?

This could eventually make robots easier and more natural to use in everyday environments. Instead of needing specialized training or manually controlling every movement, a person could simply tell Spot where to go and let the system translate that request into instructions the robot can understand.